# Daniele Artico, Ph.D.

**Postdoctoral researcher in theoretical physics**

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INSPIRE-HEP profile
personal website

Defect CFT and conformal bootstrap • Wilson lines and defect RG flows
Thermal boundaries and defects • Representation theory and operator spectra • Feynman integrals

## Research Profile

I am a postdoctoral researcher in theoretical physics at Università di Parma, working on defect conformal field theory and perturbative quantum field theory. My research combines representation theory, conformal bootstrap and Feynman-integral methods to determine operator spectra, correlation functions and defect renormalisation-group flows. Recent projects concern Wilson lines in general representations, boundaries and defects at finite temperature, and representation-theoretic constraints on the $O(n)$ spectrum. My earlier work developed integration-by-parts identities and differential equations for Feynman integrals in parameter space.

## Current Position

2025 – 2027 **Postdoctoral researcher at Università degli Studi di Parma**
PRIN Project *Constraining strongly coupled quantum field theories using symmetry*.

## Education

2020 – 2025 **Ph.D. in Theoretical Physics, Humboldt-Universität zu Berlin**
Supervisors: Prof. Dr. Jan Plefka, Prof. Dr. Dirk Kreimer

2017 – 2020 **M.Sc in Physics, Università di Torino**

2014 – 2017 **B.Sc in Physics, Università di Torino**.

2014 – 2021 **Excellence Diploma, Scuola di Studi Superiori "Ferdinando Rossi" di Torino**

## Research Publications

### Articles and Preprints

- **D. Artico** and A. Chalabi, "Boundaries and defects in conformal field theories at finite temperature," Sep. 2026. arXiv: 2609.38130 [hep-th].
- **D. Artico** and J. Roosmale Nepveu, "Constraints on the $O(n)$ model from a negative number of flavors," Aug. 2026. arXiv: 2608.12481 [hep-th].
- **D. Artico**, C. Meneghelli, M. Savi, and R. Treilis, "New Exotic Operators in the Spectrum of Wilson Lines in General Representations," Jun. 2026. arXiv: 2606.07738 [hep-th].
- **D. Artico**, J. Barrat, and Y. Xu, "Perturbative bootstrap of the Wilson-line defect CFT: Bulk-defect-defect correlators," *JHEP*, vol. 03, p. 191, Mar. 2025. DOI: 10.1007/JHEP03(2025)191 arXiv: 2410.08273 [hep-th].
- **D. Artico**, J. Barrat, and G. Peveri, "Perturbative bootstrap of the Wilson-line defect CFT: multipoint correlators," *JHEP*, vol. 02, p. 190, Feb. 2025. DOI: 10.1007/JHEP02(2025)190 arXiv: 2410.08271 [hep-th].
- **D. Artico** and L. Magnea, "Integration-by-parts identities and differential equations for parametrised Feynman integrals," *JHEP*, vol. 03, p. 096, Mar. 2024. DOI: 10.1007/JHEP03(2024)096 arXiv: 2310.03939 [hep-ph].

- **D. Artico**, S. Durham, L. Horn, F. Mezzenzana, M. Morrison, and A. Norberg, "«Beyond being analysts of doom»: Scientists on the frontlines of climate action," *Frontiers in Sustainability*, vol. 4, 2023, ISSN: 2673-4524. DOI: 10.3389/frsus.2023.1155897

### Conference Proceedings

- **D. Artico** and L. Magnea, "IBPs and differential equations in parameter space," in *Proceedings of Science (RADCOR2023)*, vol. RADCOR2023, 2024, p. 058. DOI: 10.22323/1.432.0058 arXiv: 2311.02457 [hep-ph].
- **D. Artico**, "Loop Tree Duality with generalized propagator powers: numerical UV subtraction for two-loop Feynman Integrals," in *Proceedings of Science (LL2024)*, vol. LL2024, 2024, p. 025. DOI: 10.22323/1.467.0025 arXiv: 2409.01313 [hep-ph].

### PhD Thesis

- **D. Artico**, "From Loop Integrals to Defect Correlators," Ph.D. dissertation, Humboldt-Universität zu Berlin, 2025. DOI: 10.18452/36059 arXiv: 2602.09226 [hep-th].

## Selected Talks

The complete list of my contributions and, whenever available, my slides, can be found at this *this link*.

### Talks

2026 **Wilson lines in general representations and defect RG flows**
SISSA (16 Dec, scheduled); Università di Milano-Bicocca (19 Nov, scheduled); University of Vienna (27 Oct); HU Berlin (12 Jun).

**Thermal boundaries and defects**
Thermal Seminars, online (15 Dec, scheduled); Technische Universität Wien (29 Oct, scheduled); CFT on Non-Trivial Geometries, HMI, Trinity College Dublin (7–9 Oct).

2024–2025 **Perturbative bootstrap of the Wilson-line defect CFT**
Isaac Newton Institute, Cambridge (Dec 2025); DESY Hamburg, ENS Paris and Niels Bohr Institute, Copenhagen (Nov 2024).

2024 **Feynman integrals: IBP identities and numerical methods**
Niels Bohr Institute, Copenhagen (Nov); Loops and Legs in Quantum Field Theory (Apr); Technische Universität München (Jan).

## Academic Service and Leadership

### Scientific Service

2022–2024 **Organiser of the DESY–HU Theory Seminars** | HU Berlin

2020–2021 **Organiser of the PhD Student Seminars** | HU Berlin

### Communication and Outreach

2023–2024 **Global Social Media and Communications Coordinator** | Scientist Rebellion
Coordinated a communications team of more than 20 people.

2021–2023 **Co-founder of Non-Standard Models**
Co-initiated a theoretical-physics YouTube channel with six other PhD students, featuring principal investigators and PhD students from RTG 2575, *Rethinking Quantum Field Theory*. The approximately ten-minute episodes combine explanations and animations for undergraduate physics students.

2022–2023 Participation in the *Long Night of Science* outreach project.

## Academic Service and Leadership (continued)

### Institutional Service and Representation

2025–2027 Member of the *Osservatorio paritetico contro la militarizzazione dell'Università* | Università di Parma

2018–2020 Student representative, Department of Physics | Università di Torino

2017–2020 Student representative, School of Science Teaching Committee | Università di Torino

## Teaching Experience

I led exercise classes for the following undergraduate and postgraduate courses at Humboldt-Universität zu Berlin. The official student feedback reports have always been extremely positive.

2022/23, 2024/25 **Introduction to Elementary Particle Physics** (winter semesters)
with Prof. Dr. Christophe Grojean

2024 **Quantum Mechanics** (summer semester)
with Prof. Dr. Peter Uwer

2023 **Theoretical Physics V: Thermodynamics** (summer semester)
with Prof. Dr. Peter Uwer

2022 **Theoretical Physics I: Mechanics and Special Relativity** (summer semester)
with Prof. Dr. Peter Uwer

## Skills

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|---|---|
| Languages | Proficient in Italian (mother tongue) and English (C1 level)<br>Good knowledge of Spanish<br>Basic knowledge of French and German |
| Symbolic computation | Mathematica and FORM: symbolic perturbative calculations and representation-dependent colour algebra. |
| Scientific programming | Python and C: numerical integration and scientific computation. |
| Misc. | LaTeX typesetting and publishing, image and video editing, group leadership - I coordinated a climate communication team of more than 20 people |

## References

The following senior scientists have agreed to act as referees on my behalf, and are available to provide information about myself if contacted at the e-mail addresses listed below.

- Prof. Dr. Carlo Meneghelli | carlo.meneghelli@unipr.it
- Prof. Dr. Valentina Forini | valentina.forini@physik.hu-berlin.de
- Prof. Dr. Lorenzo Magnea | lorenzo.magnea@unito.it